INCH-POUND
MIL-DTL-24643/5G
1 October 2009
SUPERSEDING
MIL-DTL-24643/5F
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +90 °C, 600 VOLTS, TYPE LSMDU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction, Watertight

First	-	Copper conductor, coated or uncoated (see table I for size).
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation (see table I for wall thickness). Standard identification code applied by Method 1 or Method 3. If identification Method 1 is used, insulation color shall be black.
Fourth	-	Nineteen conductors cabled together in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence to be consecutive, starting with no. 1 from the center outward. Fillers may be used as necessary to form a firm well-rounded assembly.
Fifth	-	An optional binder.
Sixth	-	Cross-linked polyolefin jacket (see table I for thickness).

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TABLE I. Details.

Military part no. M24643/5	Type and size	Conductor AWG size	Insulation thickness min. avg. (inch)	Cable jacket thickness min. avg. (inch)	Overall diameter max. (inch)	Conductor resistance (ohms)	Insulation resistance (megohms)
-01UN	LSMDU-6	12 (Class B)	0.028	0.060	1.000	1.730	100
-02UN	LSMDU-14	9 (Class B)	0.040	0.060	1.395	0.868	100
-03UN	LSMDU-23	7 (Class B)	0.052	0.075	1.765	0.598	100
-04UN	LSMDU-40	4 (Class C)	0.052	0.075	2.040	0.273	90
-05UN	LSMDU-60	2 (Class D)	0.052	0.075	2.330	0.172	75

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2500
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	See table I
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure

Group B:

Cold bending cable	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200 °C)	50
Drip (95±1 °C)	Zero
Physicals (unaged)	
Insulation (extruded)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	

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Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) air oven	
Insulation (extruded)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (insulation, Method 1 only) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability	No failure
Shrinkage	No failure
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Aging and compatibility (cable) (125±5 °C)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	

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Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSMDU- 6, 14 and 23	1000
LSMDU- 40 and 60	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-010)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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